ReadAloud

Oral reading fluency assessment that runs entirely on the teacher's laptop.

A child reads a passage aloud for one minute. Whisper transcribes it **on the device**, a sequence-alignment engine compares what was heard against the passage the child was supposed to read, and every deviation is classified as a specific miscue. The teacher gets a running record: words correct per minute, accuracy, instructional level, percentile band, and a printable one-pager.

> The speech model is not the product. Anyone can run Whisper.
> **The product is the scoring layer — and specifically its restraint.**
>
> Four filters that refuse to call a deviation an error when the evidence does not support it. Each one named, counted, disclosed, and overridable in both directions.

The problem it solves

A running record is the most informative minute in elementary literacy assessment, and the most expensive to collect. A teacher sits one-to-one with each child, three times a year, hand-marking every error in real time while the rest of the class does something else.

The tallying is tedious, the arithmetic is error-prone, and scoring drifts between one adult and the next — so the same child can be recorded differently depending on who held the pencil.

ReadAloud removes the tallying and applies identical rules every time. Because nothing leaves the laptop, a district does not have to ship recordings of children's voices to a vendor in order to get automated scoring.

Who it is for: K–6 classroom teachers, reading specialists, and any school operating under a universal early-literacy screening mandate but without funded staff time to meet it.

The wow factor

It reports a negative result about its own headline feature.

The dialect-sensitive suppression layer — built so a child speaking African American English is not scored down for pronunciation — **fires zero times.** We measured it. Whisper normalises the features into standard orthography (`aksed` → `asked`, `wif` → `with`) before our rules ever see them. We published the zero, explained the mechanism, and shrank the claim to the part that survived.

Every submission claims its equity feature works. This one measured whether its own does, found that it does not, and said so — then reported the *larger* finding the measurement surfaced: Whisper mis-segments dialect speech at word boundaries (`dis col` → `Diskal`), which costs a dialect-speaking reader real errors that no phonological rule can reach.

Three more things that survived adversarial review

- **116 real-word pairs were being forgiven** by the suppression stack — `house` / `horse`, `her` / `he`, `bad` / `bat`. A child who could not decode *house* was scored 100% correct and never referred. All closed, with a single gate above all four filters and regression tests for each.

- **Two of six commonly-quoted norm figures were from the wrong edition.** Grade 4 spring is 133 and grade 6 is 146 in the 2017 revision; the 1996-era values still circulate widely. Caught by checking the primary source digit-by-digit.
- **A live defect where `showed` → `said` scored as correct.** Soundex ignores vowels, so a comprehension-breaking substitution was being erased. Fixing it raised substitution recall from 80% to 90% *with zero new false positives*.

Measured results

0.48%	0.4	77
FALSE POSITIVES, CLEAN READS	MEAN WCPM ERROR	AUTOMATED TESTS, ALL GREEN

MISCUE TYPE	PRECISION	RECALL	F1
Substitution	81.8%	90.0%	85.7%
Omission	93.8%	100.0%	96.8%
Insertion	83.3%	100.0%	90.9%
Self-correction	100.0%	40.0%	57.1%

Five passages, grades 2–4, fifty planted miscues. Ground truth is **constructed, not annotated**: the spoken text is generated from the reference by applying an explicit error script, so the correct answer is known with certainty. A detection counts only if it flags the same position with the same type. **The false-positive rate is the number we optimised** — telling a child who read correctly that they made a mistake is a worse failure than missing an error.

How to demo it — three minutes

0:00 **Turn the wifi off. On stage, visibly. Then keep going.** Verified in a real browser with every external host blocked: full record → transcribe → score → print, zero outbound requests. This is the most memorable twenty seconds available to you.

0:20 **Hand the microphone to a judge.** "Would you read this paragraph out loud for me?" Then say nothing at all. Let the screen do the work.

1:20 **Point at amber.** Green is correct, red is a miscue, amber is *"we are not confident enough to call this an error, so we do not."* Click one amber word → **Count as an error.** Every number recomputes live. The line: *the teacher is the assessor, we are the stopwatch — and the override runs both ways.*

2:00 **Click "Show the student."** Same read, same words, child's language: *"You read 39 of 40 words just right."* No percentile, no instructional level, no "frustration." The child is finally addressed rather than only measured.

2:20 **Show the zero.** "Our dialect layer fired zero times across the whole evaluation. We measured it, and we published the zero."

2:45 **Hand them the printed report.** Physical paper at a software demo is disproportionately memorable. Then stop talking.

> **Insurance:** `Ctrl` + `Shift` + `D` pushes a bundled recording through the *real* pipeline — same decode, same model, same scorer. If the microphone dies or the hall is too loud, the demo continues and nothing is faked.
> **Before you start:** `lsof -ti :5174 | xargs kill` — a stale dev server will otherwise block the port.

Everything it does

Assessment engine

- **Local speech recognition** — Whisper `base.en` in a Web Worker via transformers.js, WASM by default (measured ~2× faster than WebGPU on Apple silicon for this workload).
- **Needleman-Wunsch global alignment** of transcript against the known passage, with diagonal tie-breaking so deviations read as substitutions rather than omission/insertion pairs — matching how a human marks a running record.
- **Five miscue types** — substitution, omission, insertion, self-correction, repetition. Repetitions and self-corrections are not errors, per running-record convention; self-corrections are counted separately as a positive signal of monitoring.
- **Four suppression filters**, most-specific-first: dialect, homophone, fuzzy, phonetic. Each records *which* filter fired and why.
- **A common-word gate above all four** — if both words are ordinary English, no approximate filter may bridge them. This is what stops a decoding error being erased.
- **Validity gates** — silence, empty passage, too-short reads, implausible rates, and Whisper repetition loops or trailing hallucinations ("Thanks for watching!") are refused rather than scored. An unscoreable read carries *no* verdict, so no UI can render one.
- **Metrics** — WCPM, accuracy, self-correction rate as a `1:N` ratio, instructional level, percentile band. WCPM excludes insertions (the norms' own procedure); accuracy includes them (running-record convention).

Equity and honesty

- **Dialect-sensitive layer** — seven auditable rules (th-stopping restricted to voiced /ð/, th-fronting, r-lessness, final devoicing, cluster reduction guarded against `-ed` / `-s`, metathesis, g-dropping guarded against monomorphemic `-ing`), plus copula absence gated to `is` / `are` only.

- **Nothing is ever suppressed silently** — every forgiven word appears in a panel with the rule that forgave it.
- **Bidirectional teacher override** — any flagged word can be marked correct, *and any suppressed word can be counted as an error.* Without the second direction, every control in the system would push the score upward.
- **Norms verified against the primary source**, reported as bands rather than point estimates. Below the published floor reports "below 10th"; grades outside the table report no band at all.

Interface

- Student select → passage select → read → results → printable teacher report.
- **Live input-level meter** during recording, so a dead microphone is visible immediately.
- **Colour-coded passage** — and colour never travels alone: every marked word also carries a glyph and an underline style, so the marking survives greyscale printing and colour-vision deficiency.
- **Student-facing view** — the same record in a child's language.
- **Printable one-page running record** for the teacher.
- **Accessibility** — dyslexia-friendly font stack (no webfont, so it works offline), three text sizes, high-contrast mode, full keyboard navigation, reduced-motion support.
- **Six original passages**, grades 2–4, written for this project.

Architecture

- No server · No API key · No database · No account · No language model anywhere
- There is no LLM in this product, so it **cannot hallucinate**. Alignment plus a rule set from the literacy assessment literature — deterministic and reproducible, which is what an assessment affecting a child's placement requires.
- Works fully offline once the model is cached — **empirically verified** with all external hosts blocked.
- Reproducible evaluation harness with committed results and a smoke test that exercises the real model.

Stated limitations

The evaluation uses synthesised speech, so **this has never been run on a child's voice** — the largest unknown in the system. Self-correction recall is 40% by construction: a purely semantic self-correction is indistinguishable from an inserted word without a language model, and there is deliberately no language model. Published norms cover grades 1–6 only. It is a screening aid and a research prototype, not a validated instrument; the path to one is an inter-rater reliability study against certified human scorers, stratified by dialect.

Sources. Hasbrouck & Tindal (2017), *An update to compiled ORF norms*, Tech. Report 1702, University of Oregon · Fuchs, Fuchs, Hosp & Jenkins (2001), *Scientific Studies of Reading* 5(3) · Labov & Baker (2010), *Applied Psycholinguistics* 31(4) · Charity, Scarborough & Griffin (2004), *Child Development* 75(5) · Goodman & Buck (1973), *The Reading Teacher* 27(1) · Koenecke et al. (2020), *PNAS* 117(14).